One hundred large prime-degree isogeny maps

The campaign independently verified 100 distinct maps: 50 consecutive edges in each public P192/P224 isogeny class. There are two distinct prime degrees, 2341 and 1117. The first maps start on the published sources; later maps start on their independently verified codomains.

The frozen protocol (PROTOCOL.md) defines map-level uniqueness and retains one-eigenline coverage PARTIAL at every step. Existing P192/10453 and P224/1471 maps and repeated replays are excluded. This meets a hundred-map target with repeated prime degrees allowed; a hundred different prime degrees remains a separate task.

Class	Degree	Maps	Construct (s)	Replay (s)	Total (s)
p192	2341	50	257.592	2466.415	2724.952
p224	1117	50	124.170	667.004	792.191

These are complete operational intervals on a shared Apple M4 Pro host, with two concurrent walks. Constructors have an 1800-second limit and 8 GiB RSS cap. Independent replay is outside that constructor cap. These measurements do not establish a matched speedup. All 100 exact kernel, codomain and rational curve-equation checks passed, alongside subgroup-order checks and 20 public scalar-transport checks per map (2000 total). Replay uses separate field and polynomial implementations on the same host.

SUMMARY.json (SUMMARY.json) lists all exact source/destination ICV1 and EC1 records, degrees and map hashes. P192 walk (evidence/p192/walk.json) and P224 walk (evidence/p224/walk.json) link every raw kernel and rational map, resource and stderr receipt, mapped public generator and independent replay. The embedded source revision was clean; both walks retain the binary digest.

The screen covered every prime from 1010 through 1000000 on both sources. Orders at most 8 selected 2 P192 and 3 P224 candidates; orders at most 64 selected 40 and 38. These are structural candidates. The generalized constructor supports checked extension degrees 2 through 16. All enabled degrees passed field-law tests, and the original degree-3/5 selections remain unchanged.

Validation and delivery

The CLI passed 62 tests plus 3 limb-field reference tests. Standalone algorithm suites passed 114 tests. The initial required root compilation failed with 643 errors. Three existing upstream type-repair commits restored compilation. The first complete runtime attempt exposed catalogue parser gaps, a serde-feature-dependent fixture, one sparse

fixture omission and a one MiB digest buffer overflowing the default test stack. Their original diagnostics are retained, and the parser, fixture reference and buffer allocation were repaired without changing map acceptance checks. An additional strict CLI Clippy diagnostic retains 35 errors from inherited/generated modules; that check is not presented as passing. The required algorithm Clippy job is checked separately. PR #1606 remains open and the release remains unpublished.

Canonical updates append the 100 verified models and mapped generators, rebuild covers and links, and refresh aliases, leaderboard rosters and browser joins. The complete route graph is in ROUTES.mmd (ROUTES.mmd). The IC boundary ledger, performance scoreboard and timeline retain their existing measurements because this round does not measure an IC improvement. Earlier frozen studies retain their historical findings.

Run the actual executable with: isogeny walk --curve p224 --ell 1117 --count 50 --out ./p224-walk. Replay it with: isogeny verify-walk --input ./p224-walk. Both arithmetic implementations and pinned public anchors are embedded. Runtime Cargo and a checkout are unnecessary. See tool instructions (../../tools/isogeny-cli/README.md), commands (COMMANDS.md) and manifest (MANIFEST.json).

The final required library suite is running. Push/publication remains gated on its result.

100 verified large prime-degree maps

50 per class; 2 distinct degrees; one of two eigenlines per edge (PARTIAL)

P192 class: prime degree 2341 on every edge

icv1-fp192-t31607402316713927207482677199-52e4af59

icv1-fp192-t31607402316713927207482677199-1263f86d

Verified edge 1 -> ... -> verified edge 50; full identities in SUMMARY.json

icv1-fp192-t31607402316713927207482677199-db628b10

P224 class: prime degree 1117 on every edge

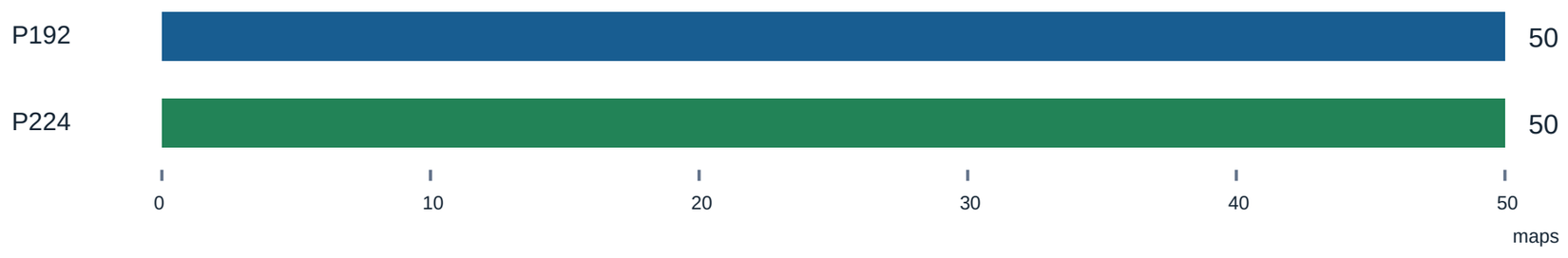
icv1-fp224-t473310010854560191642182734393082 1-fec01e99

icv1-fp224-t473310010854560191642182734393082 1-a120b97a

Verified edge 1 -> ... -> verified edge 50; full identities in SUMMARY.json

icv1-fp224-t473310010854560191642182734393082 1-a4d1f195

New distinct maps independently replayed (count)



Population: 100 maps, two public curve classes; exact checks, no sampling uncertainty.

Kernel + exact rational identity + subgroup preservation + 2000 public scalar checks: PASS